

# Kaiqi Han

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## EDUCATION

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|-----------------------------------------------------------------|---------------------|
| University of California, San Diego                             | San Diego, CA       |
| Master of Science in Computer Science                           | Sep 2025 – Jun 2027 |
| University of California, Irvine   GPA: 3.9/4.0                 | Irvine, CA          |
| Exchange Student in Electrical Engineering and Computer Science | Sep 2024 – Jun 2025 |
| Dalian University of Technology   GPA: 3.8/4.0                  | Dalian, China       |
| Bachelor of Science in Computer Science                         | Sep 2021 – Jun 2024 |

## SKILLS

**Programming Languages:** Python, C, C++, SQL, Java, JavaScript  
**Backend & Systems:** Spring Boot, FastAPI, RESTful APIs, Django, Flask, Node.js, Express, Kafka, Object-Oriented Design  
**Frontend & UI:** React, Next.js, Vue.js, HTML/CSS  
**Databases & Caching:** PostgreSQL, MySQL, MongoDB, Redis  
**Cloud & DevOps:** AWS (EC2, Lambda, S3), Docker, Kubernetes, CI/CD, Git, Linux  
**ML:** PyTorch, TensorFlow, Hugging Face, Scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, GenAI applications

## SELECTED PROJECTS

**Informal Settlement Detection & Mapping** | Python, FastAPI, PostgreSQL, Next.js, AWS

- Built a **scalable cloud-native microservices-based system** for large-scale geospatial mapping and settlement detection
- Developed fault-tolerant **FastAPI** services in a distributed environment supporting concurrent inference requests
- Designed **Next.js** web interfaces improving visualization speed for real-time mapping enabling rapid settlement analysis
- Owned full SDLC from design to **AWS** deployment with **Docker** and **CI/CD**, reducing deployment time by 50%
- Integrated **Redis** caching and monitoring mechanisms improving system reliability and reducing database load by 40%
- Stored geospatial data in **PostgreSQL** using spatial indexing enabling location-based queries for analysis purposes

**Learning Material-Sharing Platform** | Spring Boot, React, MySQL, RESTful APIs

- Built a full stack learning resource platform using **Java**, **Spring Boot**, **MyBatis** managing materials effectively
- Designed and implemented UI interaction logic using **React**, **HTML/CSS**, **JavaScript** improving user experience
- Implemented **RESTful APIs** using **JWT** authentication middleware ensuring secure access to resources reliably
- Built scalable backend architecture supporting 500+ concurrent users through efficient data handling and caching
- Deployed services on **AWS Lambda** using serverless architecture reducing deployment time by 35%
- Followed **Agile/Scrum** methodology conducting sprint planning and retrospectives delivering features iteratively

**Real-Time Task Management Platform** | Spring Boot, React, WebSocket, MySQL, Docker

- Built task management platform using **Java** and **Spring Boot** supporting 5K+ daily tasks providing real-time updates
- Implemented **WebSocket** and **Redis** pub/sub achieving sub-150ms latency for task synchronization under stress testing
- Designed **MySQL** schema for 50K+ task records with indexing, reducing query time by 60% overall across operations
- Deployed containerized services with **Docker** and **Kubernetes** achieving 99.7% in a resilient distributed environment
- Integrated **Kafka** message queue and **NoSQL** caching for asynchronous task processing reducing latency by 30%
- Configured **CI/CD** and **Git version control**, contributing to peer code reviews and reducing release cycles by 60%

**Recommender Ranking System** | Python, Spark, XGBoost

- Designed data structures and algorithm development for a distributed ranking system (200K+ interactions)
- Built distributed data preprocessing and validation pipelines with **Pandas** and **Spark** raising valid sample ratio to 95%
- Modeled implicit and explicit user feedback using **XGBoost**, achieving 0.80+ test accuracy and ~1.41 MSE
- Implemented an end-to-end training and evaluation workflow with reproducible dataset splits and experiment tracking

## WORK EXPERIENCE

**DLUT Institute of Geometric Computing and Intelligent Media** | Jan 2024 – Aug 2025

Software Engineer | Prof. Risheng Liu

**Turbulence-Aware Infrared Video Enhancement** | May 2025 – Aug 2025

- Designed a scalable distributed system for large-scale video model training (40K+ frames) using modular architecture
- Integrated pretrained components into a unified, reusable training framework to support scalable experimentation
- Implemented **PostgreSQL**-backed experiment logging with indexed queries for structured performance tracking
- Optimized **multi-GPU** training workflows and troubleshooting performance bottlenecks, reducing runtime by 40%

**Infrared Image Super-Resolution System** | Aug 2024 – Nov 2024

- Built a **PyTorch** framework supporting 500K iterations with configurable components adaptable to embedded systems
- Implemented sigmoid-gated cross-layer feature propagation to preserve high-frequency details across stages
- Optimized deformable attention modules, reducing compute cost by ~12% and improving training throughput
- Implemented automated checkpointing and managed experiment artifacts in **AWS S3** for reproducibility